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BIONEXIS

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ABSTRACT

Population growth is one of the greatest challenges facing humanity in Egypt today. As the world's population continues to increase rapidly, pressure on natural resources, food supply, and water availability increases. Our goal is to make a system that can be a smart farm to provide resources that are sufficient for the large population. Population growth rate now: about **1.4% per year. As** a result, the population reached about 120 million. Despite that Egypt produces **about 50%–60%** of its total food needs, which compared to the annual American food production, as American wheat exports were **707 million bushels** in 2023/24. Smart farm was constructed by assembling sensors (PH, turbidity, water level sensor, and soil moisture) and calibrating it then after that the prototype work by the calibrated sensors give the signals of the change happen like (the pH of the water) to the microcontroller then it gives signal to the actuator to do the function that can regulate the change which caused the change detected by the sensors. After testing, the contraption was deemed to be power efficient, effective regarding the reproduction cycles of the organism, it can increase their population effectively, and is easy to replicate, as it was only constructed using recyclable materials, such as Tupperware, empty plastic bottles, and easy-to-get electronics.

INTRODUCTION

Several Grand Challenges face Egypt, posing threats to food security and sustainability. Amongst these are population growth, which increases the demand for food production, reducing the availability of water and arable land. *"Aquaponics consists of the combination of hydroponics and aquaculture within a closed loop, being a promising technology for food production and wastewater treatment in the context of the circular economy. This technology is less energy-intensive, environmentally friendly, and consumes less water."* **Espada Sanjurjo et al. (2026)**. This emphasizes how aquaponics recycles nutrients and conserves water. Conventional farming techniques increase pollution in the water, air, and soil through fertilizers. Furthermore, the accumulation of waste along with the requirement to expand agriculture strains an already difficult situation. Specifically in Alexandria, urban issues and climatic problems exacerbate these challenges. The problem is defined by the current inability to create cost-effective automated farms using waste resources. **"**Some integrated agri-aquaculture farms using aquaponics technology can reduce water consumption by 90% compared to purely traditional agricultural farms**." Food and Agriculture Organization — 09 April 2018 — (en)**, this addresses water scarcity challenges. Each solution has strengths and flaws. Gaza's Rooftop Aquaponics Pilot made good use of recycled plastics to conserve water while increasing food availability; it lacked reliable electricity and the requirement for specialized training. The Nelson and Pade aquaponics systems in the United States provided high yields while minimizing the use of chemicals, but the construction costs precluded their application in impoverished regions. Japan's Techno Farm provides excellent automation and recycling; however, its high price and complexity prevent it from being applicable in our scope of utilizing waste resources. The Capstone challenge called for a prototype that solves these problems: it should be able to work autonomously for six continuous hours, monitor at least three parameters out of those three (temperature, pH, and salinity), use at least two actuators within the framework of a closed-loop feedback system, log data, perform variation tests, and finally, be portable. The chosen solution involves a prototype of a hybrid aquaponics system involving prawns and mint, which addresses several grand challenges, satisfying all requirements. Plastic containers will act as the tank and grow bed, providing direct aid to waste recycling. Mint and Eruca Sativa thrive in controlled environments, and prawns, as well as the shrimp database, prove useful in small farms. pH, turbidity, moisture, and water level are monitored by sensors, whereas actuators such as acid and base dosing pumps, circulation pump, and turbidity filter act via feedback loops.

MATERIALS AND METHODS

Plastic box	It will contain all of these		Turbidity sensor		To detect the clarity of the water of the fish tank
Jumber wires	To connect all parts		Water level sensor		To detect the level of water in the fish tank
Pumps	To take water from fish tank to plant		Ph sensor		To detect the PH level in the fish tank
Filter pumps	To filter the fish tank		Moisture level sensor		To detect the level of water in the plant
Relays	To control the power		Mint plant		The plant that we used
Stepdown volt	To take high volt and reduce it		Fish		The fish that we used
Breadboard	To connect all parts		Arduino		The mind of our prototype

Table 1: Materials

❖ Methods

The construction of the prototype (BioNexis) requires achieving design requirements and configuring the electric circuit. First, we make a schematic diagram of the project to simulate the prototype and its electrical connections

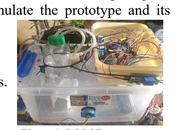


Figure 1: Initial Prototype

❖ Steps of installation:

- Firstly, a deep search into the kinds of fish and plants.
- An 8-liter plastic box was selected for the fish and another box for plants.
- Research was done about the sensors to be used, like pH sensors, turbidity sensors, water level sensors, and moisture sensors.
- After these sensors were located, they were calibrated.
- Research was done to know how to connect the fish tank to plants with pump to supply the plants with water and nutrients.
- A filter pump was used to clean the fish tank.
- Relay modules were used to control the electrical components, acting as switches.
- A deep search on how to power the prototype without batteries, and a step-down voltage regulator solved this problem.
- Jumper wires were used to connect these components to the breadboard.
- After a thorough search of the microcontrollers, Arduino was found to be the best option; it is the heart of the prototype. It is controlled by programming, in addition to the code being in AUTOSAR format, enabling the project to be compatible with bigger systems in the future.
- Finally, all components were connected to the plastic, and the prototype was ready for testing.



Figure 2: Final Prototype Close-up

❖ Methods of Testing the Prototype

A structured test plan was made to evaluate the results, which are (PH, moisture, water level, and turbidity). Each h test was conducted for 6 continuous hours over the course of 5 days, using sensors, actuators, and a microcontroller.

- **First: PH sensor output:**

As a part of our test plan, after the PH sensor was calibrated using buffer solutions of an know pH (3, 7, 11), after the rod was dunked in each of the buffers, the built-in potentiometer was twisted to match each of the corresponding pHs.

- **Second, soil moisture sensor output:**

To start, the soil moisture sensor sends signals at a baud rate of 9600, which, in contrast to the rest of the components that operate at 115200, means that all upcoming results were displayed as sufficient or insufficient rather than in numerical values.

- **Third, the water level sensor:**

As the water level sensor evaluates the water level using its exposed metal strips, combining this with the dimensions of the container (i.e, length and width), this will result in a continuous array that provides an accurate estimation of the amount of water in liters. If the water is less than 7L, a 12V water pump attached to a secondary tank resembling an external water source will refill the main tank.

Fourth, the turbidity sensor:

Turbidity is defined as Turbidity as a parameter is defined as water cloudiness by detecting how suspended particles scatter light at a 90-degree angle, which is referred to as a Nephelometric Turbidity Unit (NTU). As the turbidity sensor provides an analog output via evaluating how turbid the water is using an IR LED and detecting how much of the ray was diverted by the suspended particles. Turbidity. If it detects a turbidity of more than 40 NTU, the water filter will activate, filtering the water from the biohazards suspended in it.

RESULTS

Negative Trial	Positive Trial
As shown in Day 4 from 10am to 3pm, the turbidity rose to **44NTU from 38NTU**, at day 4 from 3pm to 8am the water level sensor deviate from **7.2L to 8.1L**, and the pH values deviated from **6.7 to 7.1** as the actuator (acid pump) was activated to oppose the change.	After solving the problems in the negative trial which the PH sensor gives sensor to the actuator then it regulate the water PH, the same implies to the water level sensor, and for the turbidity sensor when it sense that water clearness reached more than 40NTU and become constant for a time the actuator doesn't receive the signal to clear the water

The results of the PH
the PH sensor wasn't giving a constant readings as shown in the table, so we made 3 trials for five days to get 6.8 ±0.5

The results of the turbidity
As the water clearness affected with living organism in the turbidity sensor was Tring to decrease the NTU as shown in the table to get average rate 31.1 ± 0.5

The results of the water level
As shown in the graph, the water was reduced and raised with small rate, which gives us an average of 7.6 ± 0.4 L

The results of the soil moisture
It sends raw ADC to the Arduino, which is translated into status data reflect an accurate estimation of the plant hydration.

Day	Trials	Time	PH	Turbidity (NTU)	Water Level (L)	Moisture Status
Day 1	T1	10 am	7.5	26	8L	Irrigated
	T2	3 pm	7.5	27	8L	Needs replenishing
	T3	8 pm	7.4	29	7.8L	Irrigated
Day 2	T1	10 am	7.2	30	7.7L	Irrigated
	T2	3 pm	7.2	34	7.6L	Needs replenishing
	T3	8 pm	7	40	7.6L	Irrigated
Day 3	T1	10 am	6.8	25	7.5L	Needs replenishing
	T2	3 pm	6.5	28	7.4L	Irrigated
	T3	8 pm	6.5	30	7.4L	Needs replenishing
Day 4	T1	10 am	6.2	38	7.3L	Irrigated
	T2	3 pm	6.1	42	7.2 L	Needs replenishing
	T3	8 pm	6.3	44	8.1L	Needs replenishing
Day 5	T1	10 am	6.5	21	8.0L	Needs replenishing
	T2	3 pm	6.7	26	7.9L	Irrigated
	T3	8 pm	7.1	27	7.9L	Irrigated

Table 2: Results

ANALYSIS

pH Sensor Dissection: The pH sensor relies on the potential difference across the outer and inner surfaces of a glass electrode to measure the pH as follows:

*"The external reference electrode (first from left to right) is currently already integrated in the glass tube of the **"combined"** pH electrode. The **glass membrane electrodes** for pH are mainly composed of a silicate matrix containing other atoms (e.g., Na, Li). In the glass membrane, the oxygen atoms of silicates are negatively charged, and this charge is compensated by **Na^+ or Li^+** ions."*(Vilasó-Cadre, et al., 2025)

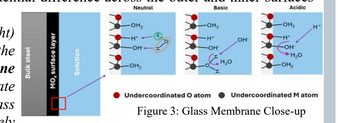


Figure 3: Glass Membrane Close-up

$$\Delta E = K + \frac{RT}{F} ln a_{H+} = k + 0.059\, log a_{H+}$$

Equation 1: Nernst equation c

The hydration increases the ion mobility, allowing an anion-exchange process with the H^+ ions of the sample, establishing a potential difference dependent on the H^+ activity in the external solution. As shown in **Figure 3**. In a potentiometric cell, several potentials are developed. There is potential associated with reference electrodes (E_{refl} **and** E_{ref2}). For a pH electrode, the **Nernst equation** describes the potential as a function of the H^+ activity, according to Equation (1). is the potential difference (**V**), ***K*** is a constant ,***R*** is the ideal gas constant (**8.314 J/mol K**), ***T*** is the temperature (K), *F* is the Faraday constant (**96,485.33 C/mol**) and is the activity of the H^+ ions.

Result analysis: After conducting the test plans mentioned previously and gathering the data from 6 hours of continuous operation across 5 days, the following was found:

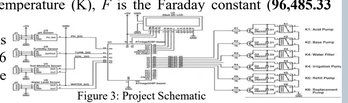
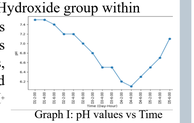


Figure 3: Project Schematic

pH Analogy: The pH of the water changes steady rate, as there is a vacant volume above the water for the prawns to be able to emerge, as they are amphibious, meaning that CO_2 is allowed to increase the acidity of the water, as $CO_2+H_2O \rightleftharpoons H_2CO_3$, then the carbonic acid partially dissociates to both $H_2CO_3 \rightleftharpoons H^+ + HCO_3^-$and $HCO_3^- \rightleftharpoons H^+ + CO_3^{2-}$ liberating at most 2 protons (H^+) from this reaction. This is opposed by the addition of $NaHCO_3$ to reduce the acidity.

In contrast, the effect of the aquatic excretions resembled in ammonia is basic upon reacting with water, as it acts as a weak base in the reaction $NH_3 + H_2O \rightleftharpoons NH_4^+ + OH^-$, which liberates a Hydroxide group within the water. This is opposed by adding NH_4Cl to the main tank. The values registered were averaging at an acceptable rate of **0.3/h** as shown in graph I. as there is a correlation between the water turning basic after the feeding times, tying them to the ammonia secretion. The rate of the **pH** shifting between acid and base was nearly **2/1** respectively, as the carbonic acid liberates at most **2** H^+ ions, while the **ammonia** liberates **only one Hydroxide group.**



Graph I: pH values vs Time

Turbidity Analogy: As the system is prone to experiencing turbidity due to the biological excretions from the aquatic prawns, a turbidity management system was installed for the purpose of keeping the **Nephelometric Turbidity Unit (NTU)** values below the threshold of **40**.

On average, the turbidity reached the threshold on **day 4** of the operation, noting that the water was replaced via the replacement pump before conducting the test. The rate of turbidity increase can be visualized as shown in graph II. On **day 4**, for example, the turbidity exceeded the limit at **6:00 pm** in T3 of the operation. After reaching the limit, the filter reduced it to a reasonable **20NTU** after 2 hours. In case the filter malfunctions or the water is too turbid **(>40NTU)**, there is an emergency mechanism that replaces the water. The replacement pump empties the tank of the turbid water, and the refill pump fills it back in about **2 minutes**.



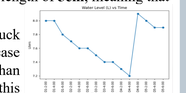
Graph II: Turbidity values vs Time

Moisture Analogy: As mentioned in the methods, the moisture sensor operates at a baud rate of 9600, whilst the rest of the components operate at **115200**, which makes the whole system prone to malfunctions during initialization if they interact with the sensor, like the LCD displaying gibberish whenever it tries fetching readings from it. So, instead of displaying numerical values, the status will be displayed. Phrases like **"needs replenishing"**, **"irrigated"**, and **"Overhydrated"** indicate how hydrated the plant is. The threshold is established in the Arduino, as it operates at different **baud rates** and **raw ADCs** simultaneously, and the actuator and the status displayed are sent from it. The plant needs only 2 minutes to be irrigated.

Water Level Analogy: As mentioned in the pH section, the water level is at 12 cm from the bottom, making the volume of water in the tank about **8.5L**, as the **length(28.5cm)** and **width(19cm)** are multiplied. This level could fluctuate according to multiple factors (e.g., temperature increase and humidity). This, however, can be controlled by the **12-9V refill pump** to maintain the level in the event of fluctuations. The effect is insignificant until the mode is changed for the database organism **(freshwater shrimp)**, as they require the whole tank to be filled **(8.5L of water).** The sensor has a conductive length of **5cm**, meaning that **5cm** needs to be submerged underwater to signal the pump to stop.

Power consumption: The project operates using a **12V** adapter attached to 2 buck converters, **9V** and **5V**, as shown in **Figure 4.** After calculating the worst-case power consumption, the result was **30 W** of total power, which is much less than your blender operating at max settings, requiring **1000W**. Moreover, using this formula, the energy efficiency is ~**93%,** as the total input is **30W** and the useful output is **28W**.

Estimated Annual Yield: Using the compound increment equation, a solid estimation of the increase in population N_0 is the initial number, **r** is the rate of reproduction, **d** is the mortality rate, **m** is the number of mating cycles, and **t** is the number of seasons. Plugging the corresponding data into the equation, it is discovered that starting with only a pair results in a **700%** increase in population.



Graph III: Water-level vs Time

$$\eta = \frac{P_{useful}}{P_{in}} = \frac{2.05}{30} \approx 0.068$$

Equation II: Power Efficiency

$$N(t) = N_0 \left[\left(1 + \frac{r}{m}\right)\left(1 - \frac{d}{m}\right)\right]^{mt}$$

Equation III: Compound Incrementation

Learning Transfer

MA.06	Allowed the deduction of relations between the parameter changing rates with respect to time, as the derivatives were introduced in this Learning outcome.
MA.10	Facilitated the calculation and laid an accurate estimation for the annual yield of the project, using the formula of compound incrementation.
BI.08	Introduced the concept of negative/positive feedback, as the human body is full of them, such as the ovarian/menstrual cycles, and the spermatogenesis, in addition to .
PH.11	Opened the possibilities of reducing/increasing the voltage and amperage, as the concepts of transformers, buck converters (step-up/down) were taught.
ME.05	Introduced the concept of power, enabling the calculation of the power consumed by the prototype and upscaling it to suit the larger module/applications.

Table 3: Learning Transfer

CONCLUSION

Five days of test results collected during the outside test confirmed that stable environmental parameters were kept in the range appropriate for prawn life and plant growth. The data obtained shows that the pH value fluctuated between 6.1-7.5 in 15 measurements, turbidity varied from 25 to 42 NTU, while the water level was kept in the range of 7.1-8.1 L. Also, irrigation of the substrate was automatically carried out whenever it was necessary due to lack of moisture. The closed loop control system worked well by providing feedback control based on sensor and actuator calibration. All Capstone project design requirements were met. There were no pauses exceeding six hours of work in a row, recycled plastic bottles were used in the prototype, and there was always a possibility of automatic regulation of environmental parameters. The nutrient recycle function was activated; thus, saving water and chemicals, which contributed to resolving some of Egypt's Grand Challenges, such as garbage recycling and waste reduction, expanding the area of agricultural lands, pollution reduction, and managing population growth and the demand for food. Thus, test results confirm that the developed inexpensive, but highly automated prototype of the aquaponic system is effective, practical, and ready to go on display. This model can demonstrate how simple feedback controls made of waste products can help create an aquaponics system in any city like Alexandria. The prototype meets all requirements to be scaled for future farming in Egypt.

RECOMMENDATION

If the project is to be replicated at an agency-level plant, some adjustments and further upgrades are a necessity to achieve the optimal outcome possible. These upgrades are:

- **Improving circulation:** Instead of using plastic tubes, metal ones will be more durable and will negate malfunctions with less maintenance cost. However, if the recycling trait is to be put in perspective, using durable PLA or PVC pipes could theoretically have the same effect, with more maintenance costs, unfortunately. Plus, adding more sensors will secure the organisms' well-being
- **Reducing External Power Reliance:** Instead of using all the power from external power infrastructure or commodities to lift water, for example, generating a part of it via attaching water turbines to the pipelines could reduce the operation cost, making the profitability much higher and a viable option for future investors.
- **The Implementation of Nitrobacter:** Aquatic creatures not only secrete ammonia and nitrates in their excretions, but also nitrites, which are poisonous for plants. Utilizing Nitrobacter to break down the nitrites in the pumping tubes or installing a nitrite filtering station will ensure the plant's well-being.

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